

POC Implementation:

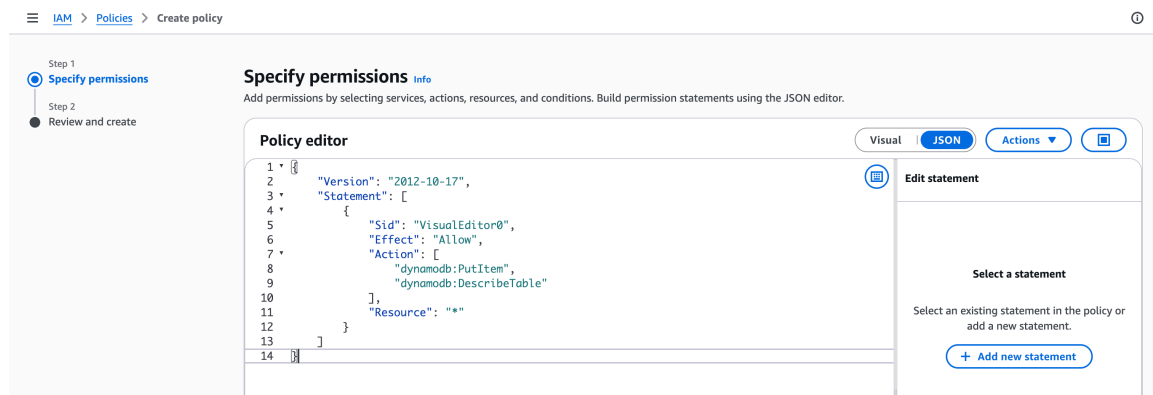
Following is the task-by-task process of creating a proof of concept with screenshots:

- Create IAM policies and roles to follow best practices of working in the AWS Cloud.
- Create a DynamoDB table to store data.
- Create an Amazon SQS queue to receive, store, and send messages between software components.
- Create Lambda functions and set up triggers to invoke actions in different AWS services.
- Enable DynamoDB Streams to capture modifications in the database table.
- Configure Amazon SNS to receive email or text notifications.
- Create a REST API to insert data into a database.

Task 1. Setup: Security & Access Management Design

Task 1.1: Creating custom IAM policies

Lambda-Write-DynamoDB



Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy name
Enter a meaningful name to identify this policy.

Description - optional
Add a short explanation for this policy.

Maximum 1,000 characters. Use alphanumeric and '+, @, _' characters.

[Edit](#)

Search

☒ Show remaining 468 services

Lambda-SNS-Publish

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Visual | **JSON** | Actions

--	--

Select an existing statement in the policy or
add a new statement.

+ Add new statement

Lambda-DynamoDBStreams-Read

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Visual

JSON

Actions

Policy editor

```
1 * {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "dynamodb:GetShardIterator",
9         "dynamodb:DescribeStream",
10        "dynamodb:ListStreams",
11        "dynamodb:GetRecords"
12      ],
13      "Resource": "*"
14    }
15  ]
16 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

Step 1

Specify permissions

Step 2

Review and create

Review and create

Info

Review the permissions, specify details, and tags.

Policy details

Policy name

Enter a meaningful name to identify this policy.

Lambda-DynamoDBStreams-Read

Maximum 128 characters. Use alphanumeric and '+', '@', '-' characters.

Description - optional

Add a short explanation for this policy.

Maximum 1,000 characters. Use alphanumeric and '+', '@', '-' characters.

Permissions defined in this policy

Info

Edit

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Allow (1 of 469 services)

Show remaining 468 services

Service	Access level	Resource	Request condition
DynamoDB	Limited: Read	All resources	None

Add tags - optional

Info

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Lambda-Read-SQS

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Visual

JSON

Actions

Policy editor

```
1 * {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "VisualEditor0",
6       "Effect": "Allow",
7       "Action": [
8         "sqs:DeleteMessage",
9         "sqs:ReceiveMessage",
10        "sqs:GetQueueAttributes",
11        "sqs:ChangeMessageVisibility"
12      ],
13      "Resource": "*"
14    }
15  ]
16 }
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

Task 1.2: Creating IAM roles and attaching policies to the roles

IAM

>

Roles

>

Create role

Step 1

●

Select trusted entity

Step 2

●

Add permissions

Step 3

○

Name, review, and create

Select trusted entity

Info

Trusted entity type

☒ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

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IAM

>

Roles

>

Create role

Step 1

●

Select trusted entity

Step 2

●

Add permissions

Step 3

○

Name, review, and create

Add permissions

Info

Permissions policies (2/1152)

Info

Choose one or more policies to attach to your new role.

Filter by Type

Search

lambda-

×

All types

▼

4 matches

< 1 >

⚙

	Policy name ⓘ	Type	Description
☐	Lambda-DynamoDBStreams-Read	Customer managed	-
☒	Lambda-Read-SQS	Customer managed	-
☐	Lambda-SNS-Publish	Customer managed	-
☒	Lambda-Write-DynamoDB	Customer managed	-

► Set permissions boundary - optional

Cancel

Previous

Next

Step 1

● Select trusted entity

Step 2

● Add permissions

Step 3

● Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+@-./\{}#%*^&~:"'

Step 1: Select trusted entities

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
```

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Step 1

● Select trusted entity

Step 2

● Add permissions

Step 3

● Name, review, and create

Role Lambda-SQS-DynamoDB created.

View role

Add permissions [info](#)

Permissions policies (2/1152) [info](#)

Choose one or more policies to attach to your new role.

Filter by Type

All types

 4 matches < 1 >

☐	Policy name ↗	Type	Description
☒	Lambda-DynamoDBStreams-Read	Customer managed	-
☐	Lambda-Read-SQS	Customer managed	-
☒	Lambda-SNS-Publish	Customer managed	-
☐	Lambda-Write-DynamoDB	Customer managed	-

▶ Set permissions boundary - optional

Cancel

Previous

Next

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IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

Lambda-DynamoDBStreams-SNS

Maximum 64 characters. Use alphanumeric and '+*=@-_' characters.

Description

Add a short explanation for this role.

Allows Lambda functions to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+*, @-/[]#%'^{};~"

Step 1: Select trusted entities

Edit

Trust policy

```
1 - {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ]
9     }
10  ]
11 }
```

IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Select trusted entity Info

Trusted entity type

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

API Gateway

Use case

☒ **API Gateway**

Use the API Gateway to push logs to CloudWatch Logs.

IAM > Roles > Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Add permissions Info

Permissions policies (1) Info

The type of role that you selected requires the following policy.

Policy name

AmazonAPIGatewayPushToCloudWatchLogs

AWS managed

Set permissions boundary - optional

Cancel

Previous

Next

IAM

>

Roles

>

Create role

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

APIGateway-SQS

Maximum 64 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description

Add a short explanation for this role.

Allows API Gateway to push logs to CloudWatch Logs.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/[\()]\#\$\$%^*{};:"'

Step 1: Select trusted entities

Edit

Trust policy

1 - {

2 - {

3 - "Version": "2012-10-17",

4 - "Statement": [

5 - {

6 - "Sid": "",

7 - "Effect": "Allow",

8 - "Principal": {

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Task 2: Creating a DynamoDB table

DynamoDB

Tables

Create table

Share your feedback on Amazon DynamoDB

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Create table

Table details

info

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

Table name

This will be used to identify your table.

orders

Between 3 and 255 characters, containing only letters, numbers, underscores (_), hyphens (-), and periods (.).

Partition key

The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

orderIdString

1 to 255 characters and case sensitive.

Sort key - optional

You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Enter the sort key nameString

1 to 255 characters and case sensitive.

Table settings

Default settings

The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose 'Customize settings'.

Customize settings

Use these advanced features to make DynamoDB work better for your needs.

Default table settings

These are the default settings for your new table. You can change some of these settings after creating the table.

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Task 3: Creating an SQS queue

Application Integration

Amazon SQS

A message queuing service

Amazon SQS provides queues for high-throughput, system-to-system messaging. You can use queues to decouple heavyweight processes and to buffer and batch work. Amazon SQS stores messages until microservices and serverless applications process them.

Get started

Learn how to use Amazon SQS by creating a queue, sending a message to the queue, and receiving and processing the message.

Create queue

Pricing

You can get started with Amazon SQS for free. All customers can make 1 million Amazon SQS requests for free each month. Some applications might be able to operate within this Free Tier limit.

Cost calculator

Documentation

Developer guide

API reference

FAQs

How it works

The diagram illustrates the Amazon SQS workflow. On the left, 'Producers' (applications, microservices, and other AWS services) send messages to the 'Amazon SQS' queue. The queue is represented as a box with a cylinder icon. On the right, 'Consumers' (applications, Lambda functions, EC2 instances, and other AWS services) receive messages from the queue. The queue itself is labeled 'SQS Queue' and 'Queues store messages and wait for consumers to poll'.

Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

Standard info

At-least-once delivery, message ordering isn't preserved

- At-least-once delivery
- Best-effort ordering

FIFO info

First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

You can't change the queue type after you create a queue.

Name

POC-Queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Amazon SQS > Queues > Create queue

Access policy

Define who can access your queue.

Choose method

Basic

Use simple criteria to define a basic access policy.

Advanced

Use a JSON object to define an advanced access policy.

Define who can send messages to the queue

Only the queue owner

Only the owner of the queue can send messages to the queue.

Only the specified AWS accounts, IAM users and roles

Only the specified AWS account IDs, IAM users and roles can send messages to the queue.

arn:aws:iam::[redacted]:role/APIGateway-SQS

Define who can receive messages from the queue

Only the queue owner

Only the owner of the queue can receive messages from the queue.

Only the specified AWS accounts, IAM users and roles

Only the specified AWS account IDs, IAM users and roles can receive messages from the queue.

arn:aws:iam::[redacted]:role/Lambda-SQS-DynamoDB

JSON (read-only)

```
{
  "Version": "2012-10-17",
  "Id": "__default_policy_ID",
  "Statement": [
    {
      "Sid": "__owner_statement",
      "Effect": "Allow",
      "Principal": {
        "AWS": "[redacted]"
      },
      "Action": [
        "SQS:*"
      ],
      "Resource": "arn:aws:sqs:us-east-2:[redacted]:POC-Queue"
    },
    {
      "Sid": "__sender_statement",
      "Effect": "Allow",

```

Redrive allow policy - Optional

Identify which source queues can use this queue as the dead-letter queue.

Select which source queues can use this queue as the dead-letter queue.

Disabled

Enabled

Task 4: Order Processing Service Implementation

Task 4.1: Creating a Lambda function for the Lambda-SQS-DynamoDB role

☰

Lambda > Functions > Create function

ⓘ

🔍

Create function Info

Choose one of the following options to create your function.

Author from scratch

Start with a simple Hello World example.

Use a blueprint

Build a Lambda application from sample code and configuration presets for common use cases.

Container image

Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

POC-Lambda-1

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime Info

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.10

Permissions

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **More settings**.

Custom settings Info

Durable execution - new

Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider - new

Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Configure custom execution role

Choose an existing execution role or create a new one.

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

Lambda-SQS-DynamoDB

Create new role

View role details in IAM ↗

Close

Save

Task 4.2: Setting up Amazon SQS as a trigger to invoke the function

☰

Lambda > Functions > POC-Lambda-1

▼ Function overview Info

Diagram

Template

POC-Lambda-1

Layers (0)

+ Add trigger

+ Add destination

Code

Test

Monitor

Configuration

Aliases

Versions

Code source Info

EXPLORER

POC-LAMBDA-1

lambda_function.py

DEPLOY

Current

Deploy (⇧⌘U)

Test (⇧⌘I)

lambda_function.py

```
1 import json
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
7         'body': json.dumps('Hello from Lambda!')
8     }
9
```

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Lambda > Add triggers

Add trigger

Trigger configuration [Info](#)

SQS

aws event-source-mapping polling queue

SQS queue

Choose or enter the ARN of an SQS queue.

Q

arn:aws:sqs:us-east-2:██████████:POC-Queue

×

↺

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Provisioned mode [Info](#)

Select to configure provisioned mode for your event source mapping, which allows you to optimize the throughput of your event-driven applications. [View pricing](#).

☐ Configure

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size - optional

The maximum number of records in each batch to send to the function.

10

The maximum is 10,000 for standard queues and 10 for FIFO queues.

Task 4.3: Adding and deploying the function code

Lambda > Functions > POC-Lambda-1

POC-Lambda-1

Layers (0)

SQS

+ Add trigger

+ Add destination

Code

Test

Monitor

Configuration

Aliases

Versions

Code source [Info](#)

← →

POC-Lambda-1

EXPLORER

POC-LAMBDA-1

lambda_function.py

DEPLOY Undeployed

Deploy (⌘⌘U)

Test (⌘⌘I)

lambda_function.py

```
1 import boto3, uuid
2
3 client = boto3.resource('dynamodb')
4 table = client.Table("orders")
5
6 def lambda_handler(event, context):
7     for record in event['Records']:
8         print("test")
9         payload = record["body"]
10        print(str(payload))
11        table.put_item(Item= {'orderId': str(uuid.uuid4()), 'order': payload})
```

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Task 4.4: Testing the POC-Lambda-1 Lambda function

Lambda > Functions > POC-Lambda-1

CodeTestMonitorConfigurationAliasesVersions

Test eventInfo

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

- Create new event
- Edit saved event

Invocation type

- Synchronous
 - Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.
- Asynchronous
 - Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event namePOC-Lambda-Test-1

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

- Private
 - This event is only available in the Lambda console and to the event creator. You can configure a total of 10. Learn more
- Shareable
 - This event is available to IAM users within the same account who have permissions to access and use shareable events. Learn more

Template - optionalSQS

Format JSON

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Event JSON

Format JSON

```
1 {  
2   "Records": [  
3     {  
4       "messageId": "19dd0b57-b21e-4ac1-bd88-01bbb068cb78",  
5       "receiptHandle": "MessageReceiptHandle",  
6       "body": "Hello from SQS!",  
7       "attributes": {  
8         "ApproximateReceiveCount": "1",  
9         "SentTimestamp": "1523232000000",  
10        "SenderId": "123456789012",  
11        "ApproximateFirstReceiveTimestamp": "1523232000001"  
12      },  
13      "messageAttributes": {},  
14      "md5OfBody": "{{[md5_of_body]}}",  
15      "eventSource": "aws:sqs",  
16      "eventSourceARN": "arn:aws:sqs:us-east-1:123456789012:MyQueue",  
17      "awsRegion": "us-east-1"  
18    }  
19  ]  
20 }
```

1:1JSON

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CodeTestMonitorConfigurationAliasesVersions

Executing function: succeeded(logs)

Details

```
{  
  "statusCode": 200,  
  "body": "\nHello from Lambda\n"}
```

Summary

Code SHA-256 HAPq9EReJVEc5gLatrcgyd5vtD9eiUGF932t0BxY=	Execution time 21 seconds ago
Function version \$LATEST	Request ID 72701659-6af5-4da6-8567-83614e65bb08
Duration 1.59 ms	Billed duration 88 ms
Resources configured 128 MB	Max memory used 31 MB
Init duration 85.52 ms	

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Task 4.5: Verifying that the Lambda function adds the test message to a database

Edit item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Attributes

Add new attribute

Attribute name	Value	Type	Remove
orderID - Partition key	e5a7cd07-cd99-431a-b8e5-187de1407f2c	String	
order	Hello from SQS!	String	Remove

Cancel

Save

Save and close

Task 5: Enabling DynamoDB Streams

DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Parameter groups

Events

Filter by tag

Any tag key

Filter by tag value

Any tag value

Find tables

orders

Indexes

Monitor

Global tables

Backups

Exports and streams

Permissions

Exports to S3

Info

Last updated

May 24, 2026, 03:40 (UTC+1:00)

View details

Export to S3

Showing all export jobs from the last 90 days.

Find exports

0 matches

Initiated by = User

Clear filters

Export ARN

Destination S3 bucket

Status

Export job start time (UTC+01:00)

Export type

No exports

Export to S3

DynamoDB stream details

Turn off

Stream status

On

View type

New image

Latest stream ARN

arn:aws:dynamodb:us-east-2:123456789012:table/orders/stream/2026-05-24T02:40:52.617

Resource-based policy

Not active

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DynamoDB > Tables > orders > Turn on DynamoDB stream

Share your feedback on Amazon DynamoDB

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Turn on DynamoDB stream

DynamoDB stream details

Capture item-level changes in your table, and push the changes to a DynamoDB stream. You then can access the change information through the DynamoDB Streams API.

View type

Choose which versions of the changed items you would like to push to the DynamoDB stream.

☐ New and old images

Both the new and old images of the changed item.

☐ Key attributes only

Only the key attributes of the changed item.

☒ New image

The entire item as it appears after it was changed.

☐ Old image

The entire item as it appears before it was changed.

Cancel

Turn on stream

Task 6: Creating an SNS topic and setting up subscriptions

Task 6.1: Creating a topic in the notification service

Amazon SNS > Topics > Create topic

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create topic

Details

Type | Info

Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

POC-Topic

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional | Info

To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

Maximum 100 characters.

► Encryption - optional

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

Task 6.2: Subscribing to email notifications

Amazon SNS > Topics > POC-Topic

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

POC-Topic

Edit Delete Publish message

Details

Name

POC-Topic

ARN

arn:aws:sns:us-east-2:123456789012:POC-Topic

Display name

-

Type

Standard

Topic owner

123456789012

Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Integrations

Subscriptions (0)

Edit Delete Request confirmation Confirm subscription Create subscription

Search

< 1 > ⚙

ID	Endpoint	Status	Protocol
No subscriptions found			
You don't have any subscriptions to this topic.			

Create subscription

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New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARN

arn:aws:sns:us-east-2:██████████:POC-Topic

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

██████████@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

Subscription to POC-Topic created successfully.
The ARN of the subscription is arn:aws:sns:us-east-2:370047118847:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e.

Notifications 0 0 1 1 0 0

Subscription: bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Edit Delete

Details

ARN
arn:aws:sns:us-east-2:██████████:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Endpoint
██████████@gmail.com

Topic
[POC-Topic](#)

Status
Pending confirmation

Protocol
EMAIL

[Subscription filter policy](#) Redrive policy (dead-letter queue)

Subscription filter policy [Info](#)

This policy filters the messages that a subscriber receives.

No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

Edit

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Subscription: bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Edit Delete

Details

ARN
arn:aws:sns:us-east-2:██████████:POC-Topic:bdaba1fa-21b1-448b-b1e8-c44c0816da0e

Endpoint
██████████@gmail.com

Topic
[POC-Topic](#)

Status
Confirmed

Protocol
EMAIL

Lambda > Add triggers

Trigger configuration info

DynamoDB

aws database event-source-mapping nosql polling

DynamoDB table

Select or enter the ARN of a DynamoDB table. To select a stream in another AWS account, enter its DynamoDB stream Amazon Resource Name (ARN).

arn:aws:dynamodb:us-east-2:123456789012:table/orders

Event poller configuration

☒ Activate trigger

Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

☐ Enable EventCount metrics

Track the number of events polled, filtered, invoked, and dropped by your event source mapping. CloudWatch charges apply.

Batch size

The maximum number of records in each batch to send to the function.

100

Starting position

The position in the stream to start reading from.

Latest

Batch window - optional

The maximum amount of time to gather records before invoking the function, in seconds.

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Task 7.3: Configuring the second Lambda function

Code source info

Open in Visual Studio Code

Update

POC-Lambda-2

lambda_function.py

```
1 import boto3, json
2
3 client = boto3.client('sns')
4
5 def lambda_handler(event, context):
6
7     for record in event["Records"]:
8
9         if record['eventName'] == 'INSERT':
10             new_record = record['dynamodb']['NewImage']
11             response = client.publish(
12                 TargetArn='<Enter Amazon SNS ARN for the POC-Topic>',
13                 Message=json.dumps({'default': json.dumps(new_record)}),
14                 MessageStructure='json'
15             )
```

EXPLORER

POC-LAMBDA-2

lambda_function.py

DEPLOY

Deploy (⌘%U)

Test (⌘%T)

TEST EVENTS [NONE SELECTED]

Create new test event

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Task 7.4: Testing the POC-Lambda-2 Lambda function

☰

Lambda > Functions > POC-Lambda-2

🔍

🔧

☑️ Successfully updated the function "POC-Lambda-2".

Code

Test

Monitor

Configuration

Aliases

Versions

Test event Info

CloudWatch Logs Live Tail

Save

Test

To invoke your function without saving an event, configure the JSON event, then choose Test.

Test event action

Create new event

Edit saved event

Invocation type

Synchronous

Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Asynchronous

Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

POC-Lambda-Test-2

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings

Private

This event is only available in the Lambda console and to the event creator. You can configure a total of 10. [Learn more](#)

Shareable

This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

DynamoDB Update

🔄

Event JSON

Format JSON

☑️ The test event "POC-Lambda-Test-2" was successfully saved.

Notifications 0 0 2 0 0 0

Code

Test

Monitor

Configuration

Aliases

Versions

☑️ Executing function: succeeded [\(logs\)](#)

▼ Details

null

Summary

Code SHA-256

DAo0Vv56wYg+P1eMzWM9QfAJ08lQytu9B8R98G/EPUg=

Function version

\$LATEST

Duration

249.46 ms

Resources configured

128 MB

Init duration

475.03 ms

Execution time

21 seconds ago

Request ID

690348a3-3d7d-4833-82ab-43a7937cd715

Billed duration

725 ms

Max memory used

79 MB

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

START RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715 Version: \$LATEST
END RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715
REPORT RequestId: 690348a3-3d7d-4833-82ab-43a7937cd715 Duration: 249.46 ms Billed Duration: 725 ms Memory Size: 128 MB Max Memory Used: 79 MB Init Duration: 475.03 ms

☑️ CloudShell

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API Gateway

Create and manage APIs at scale

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs.

Get started

Create a new API to begin exploring API Gateway. You can also import an external definition file into API Gateway.

Create an API

How it works



API Gateway enables you to connect and access data, business logic, and functionality from backend services such as workloads running on Amazon Elastic Compute Cloud (Amazon EC2), code running on AWS Lambda, any web application, or real-time communication applications.

Pricing

With Amazon API Gateway, you only pay when your APIs are in use. There are no minimum fees or upfront commitments. For HTTP and REST APIs, you pay based on API calls received and amount of data transferred out. For WebSocket APIs, you pay based on number of messages and connection duration.

[Learn more about pricing](#)

Resources

- [Getting Started Guide](#)
- [Developer Guide](#)
- [API References](#)

API Gateway > APIs > Create API > Create REST API

Create REST API

API details

☒ New API
Create a new REST API.

☐ Clone existing API
Create a copy of an API in this AWS account.

☐ Import API
Import an API from an OpenAPI definition.

☐ Example API
Learn about API Gateway with an example API.

API name
POC-API

Description - optional

API endpoint type
Regional APIs are deployed in the current AWS Region. Edge-optimized APIs route requests to the nearest CloudFront Point of Presence. Private APIs are only accessible from VPCs.
Regional

Security policy - new | Info
Transport Layer Security (TLS) protects data in transit between a client and server. The security policy also determines the cipher suite options that clients can use with your API.
Choose a security policy

IP address type | Info
Select the type of IP addresses that can invoke the default endpoint for your API.
☒ IPv4
Supports only edge-optimized and Regional API endpoint types.
☐ Dualstack

Method details

Method type
POST

Integration type

☐ Lambda function
Integrate your API with a Lambda function.



☐ HTTP
Integrate with an existing HTTP endpoint.



☐ Mock
Generate a response based on API Gateway mappings and transformations.



☒ AWS service
Integrate with an AWS Service.



☐ VPC link
Integrate with a resource that isn't accessible over the public internet.



AWS Region
us-east-2

AWS service
Simple Queue Service (SQS)

AWS subdomain

HTTP method
POST

Action type

Task 9: Testing the architecture by using API Gateway

API Gateway

APIs

Custom domain names

Domain name access associations

VPC links

AgentCore targets [New](#)

▼ API: POC-API

Resources

Stages

Authorizers

Gateway responses

Models

Resource policy

Documentation

Dashboard

API settings

Usage plans

API keys

Client certificates

Settings

Create resource

/

POST

Client

Method request

Integration request

Method response

Integration response

AWS integration

Method request

Integration request

Integration response

Method response

Test

Test method

Make a test call to your method. When you make a test call, API Gateway skips authorization and directly invokes your method.

Query strings

param1=value1¶m2=value2

Headers

Enter a header name and value separated by a colon (:). Use a new line for each header.

header1:value1

header2:value2

Client certificate

No client certificates have been generated.

Request body

1 { "item": "latex gloves",

2 "customerID": "12345" }

CloudShell

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DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

▼ DAX

Clusters

Subnet groups

Parameter groups

Events

Tables (1)

Filter by tag

Any tag key

Filter by tag value

Any tag value

Find tables

1

orders

orders

Scan or query items

Scan

Query

Select table or index

Table - orders

Select attribute projection

All attributes

Filters - optional

Run

Reset

Completed · Items returned: 2 · Items scanned: 2 · Efficiency: 100% · RCUs consumed: 2

Table: orders - Items returned (2)

Scan started on May 24, 2026, 04:22:26

orderID (String) | order

e5a7cd07-cd99-431a... Hello from SQS!

62c6ecde-72aa-4e88-... { "item": "latex gloves", "customerID": "12345" }

CloudShell

Feedback

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DynamoDB

Explore items: orders

Edit item

Share your feedback on Amazon DynamoDB

Your feedback is an important part of helping us provide a better customer experience. Take this short survey to let us know how we're doing.

Share feedback

Edit item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form | JSON view

Attributes

Add new attribute

Attribute name

Value

Type

Remove

orderID - Partition key

62c6ecde-72aa-4e88-851e-69ffebc3646f

String

order

{ "item": "latex gloves", "customerID": "12345" }

String

Remove

Cancel

Save

Save and close